New words are assumed to have specific and atypical meanings

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Consider two loanwords whose meanings shifted when borrowed into a new language: *salsa* means any type of sauce in Spanish, but when borrowed into English came to mean specific tomato- and pepper-based sauces; *chai* means any type of tea in Hindi, but when borrowed into English came to mean a specific type of spiced, milky tea. What explains this shift?

We propose that semantic narrowing in language contact may reflect the accumulation of individual-level contrastive inferences: when speakers encounter a new word for an already nameable concept, they assume it must refer to something more specific. People avoid having multiple language forms that correspond to the exact same meanings (Clark, 1990, 2014). Indeed, language documentation and experimental tasks show that people avoid absolute synonymy (Altenhof & Roberts, 2023; Geeraerts, 2009; Markman & Wachtel, 1988; Andersen, 2014). As a result, when encountering a new word, speakers must differentiate its meaning from meanings of known words.

The process of accommodating word meanings into an already populated conceptual space is also the process of word learning generally. Several established principles of word learning provide a foundation for the narrowing inference we investigate. All else equal, children and adults tend to assume that words refer to basic-level categories (e.g., *leaf, shoe*) (Rosch, 1975; Markman, 1989; Golinkoff, Mervis, & Hirsh-Pasek, 1994). How, then, do they learn concepts at more specific levels? Children and adults assume that speakers sample across the breadth of a category (*strong sampling*), such that when shown three dalmatians labeled with a novel word, they infer that the word applies to a subordinate category (dalmatians), not the basic-level one (dogs) (Xu & Tenenbaum, 2007). Adults and children also infer that words applied to atypical exemplars (e.g., a blowfish compared to a typical goldfish) refer to subordinate categories (Emberson, Loncar, Mazzei, Treves, & Goldberg, 2019). These inferential processes emerge in the preschool age range, a developmental period of rapid vocabulary growth.

Here, we investigate people's inferences about novel words' meanings in controlled word learning experiments, taking advantage of people's tendencies to

sample across the breadth of a category and associate atypical exemplars with subordinate categories. We present two experiments with adults and 3- to 5-year-old children testing whether people semantically narrow the meanings of new words that overlap with existing vocabulary. This tendency, if present, would shape children's word learning in early childhood, adults' accommodation of new vocabulary, and, over the course of many individual-level inferences, the semantic narrowing of borrowed words.

Method. Adult participants (N = 60) were recruited through Prolific for a web experiment. Child participants (N = 48; 18 3-year-olds, 18 4-year-olds, and 12 5-year-olds) were recruited at a preschool and participated using a tablet. The adult and child tasks were nearly identical.

Each participant was assigned to the familiar word or the novel word condition. On each trial, participants saw a grid of 12 objects, all belonging to the same category (e.g., leaves), equally divided among four subtypes (e.g., bean-shaped, teardrop-shaped, oak-shaped, and heart-shaped leaves). Two subtypes were more atypical and two were more typical, as designated a priori by experimenters (e.g., teardrop-shaped leaves were typical, and bean-shaped leaves were atypical). In a post-hoc typicality norming experiment with 31 independent raters, these subtypes were verified to effectively manipulate typicality (on a scale of 0 to 100, *mean typical* = 72.56 compared to *mean atypical* = 41.22, $p < 0.001$ by t-test).

Adult participants were asked to, e.g., "Select two [leaves/nefts]" via text; child participants were asked to, e.g., "Tap on two [leaves/nefts]" via audio. Participants clicked (for adults) or tapped (for children) on the images to respond. Each participant saw four trials, one for each category (leaves, shells, mushrooms, and flowers). Children completed two practice trials with familiar referents at the beginning of the experiment to ensure they understood the task.

Results. We predicted that participants would choose (1) more specific subtypes and (2) more atypical referents when prompted with an unfamiliar word compared to the familiar word for a given category. Both typicality of choices (typical or atypical) and specificity of choices (two items of the same subtype or two items of different subtypes) were modeled as a function of condition (novel vs. familiar) using generalized linear mixed-effects models with a binomial link; all models included by-subject and by-category random intercepts, and the child models included a mean-centered age predictor.

Both adults and children choose more atypical items when prompted with a novel word compared to a familiar one (26% for familiar vs. 60% for novel among adults, $\beta_{adult} = 1.62, p < 0.001$; 34% for familiar vs. 53% for novel among children, $\beta_{child} = 1.03, p = 0.009$). Children and adults also choose more specific referents (i.e., are less likely to choose items of different subtypes) when prompted with a novel word compared to a familiar word (57% choosing two subtypes for familiar vs. 24% for novel among adults, $\beta_{adult} = -3.01, p = 0.001$; 37% for familiar vs. 11% for novel among children, $\beta_{child} = -1.71, p = 0.006$) (Fig. 1).

We do not find effects of or interactions with age.

In sum, both adults and 3- to 5-year-olds selected items consistent with inferring that novel words meant something more specific and atypical than words already in their vocabulary. This finding goes beyond existing evidence that people assume novel words refer to novel objects, instead showing that people infer novel words refer to specific and atypical subsets of *familiar* concepts. These inferences, when repeated over people and time, may contribute to the semantic narrowing of borrowed words' meanings, bridging individuals' inferences with diachronic change.

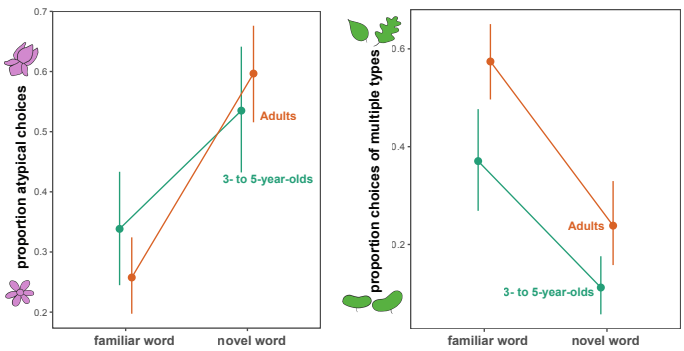

Figure 1. Adults and 3- to 5-year-old children choose more atypical and more specific selections of exemplars when prompted with novel words.

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